

Vaccination of the Egg-Allergic Individual

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Summary

- Egg allergic individuals may be safely vaccinated with the **measles mumps rubella (MMR)** vaccine, the **measles mumps rubella varicella (MMR-V)** vaccine (which contains no egg protein) and the **influenza** vaccine, including both inactivated (injectable) and live attenuated (intranasal) influenza vaccines, which may contain minute traces of egg protein.
- Contemporary influenza vaccines used in Australia and New Zealand contain very low levels of residual ovalbumin, generally less than or equal to 1 microgram per dose, a quantity substantially below estimated oral reaction thresholds in egg-allergic individuals. Prospective and retrospective studies involving egg-allergic individuals (including those with a prior history anaphylaxis to egg) receiving injectable inactivated, and intranasal live attenuated influenza vaccines have reported no vaccine-induced anaphylaxis.
- Special precautions such as split dosing, prior allergy testing with the vaccines, allergy specialist review before vaccination or prolonged waiting times after administration are not required.
- The **yellow fever** and **Q fever** vaccines potentially contain higher amounts of egg protein and allergy specialist evaluation is recommended before vaccination.

Introduction

Vaccination is an important method of reducing the risk of developing a number of infectious diseases. Before mass vaccination campaigns in Australia, **measles** was a significant contributor to hospitalization, morbidity (such as pneumonia, meningitis and encephalitis) and sometimes death www.health.gov.au/internet/main/publishing.nsf/content/cda-cdi3901a.htm.

Influenza is also a significant cause of morbidity and mortality in Australia and New Zealand. In 2025 there were over 500,000 confirmed cases (1) and 1,738 deaths in Australia due to influenza, including 22 deaths in young people (<20 years old) in 2024-25 (2). In 2024, influenza-associated severe acute respiratory illness (SARI) hospitalisation in New Zealand reached its highest cumulative incidence since 2018, with 564 influenza-positive SARI cases identified (3). Notification and hospitalisation rates related to influenza infection are highest in children aged 0-5 years (the age group most affected by egg allergy 4, 5) and in adults aged 70+ years www.health.gov.au/flureport.

Available influenza vaccines include egg protein-free influenza vaccines grown in cell lines,, and inactivated influenza vaccines (IIV) and live attenuated influenza vaccines (LAIV) which are derived from influenza virus grown in hen's egg. Thus, egg-based inactivated and live attenuated influenza vaccines potentially may contain minute traces of egg protein. The ability to safely vaccinate egg-allergic individuals (particularly in the context of potentially pandemic infection) is an important public health issue.

Older vaccination guidelines and the vaccine Product Information (PI) recommended avoidance of influenza vaccination in egg-allergic individuals based on case reports of anaphylaxis 30 years ago, when the amounts of egg protein were much higher than currently (6, 7). A number of studies over the last ten years have shown no greater risk of influenza vaccine allergy in those with or without egg allergy (8). This is not surprising given that the amount of egg ovalbumin present in Australian and New Zealand vaccines is currently ~ 1 microgram (microg) or less per dose, substantially less than the estimated 130 microg egg protein taken orally considered likely to trigger reactions in egg allergic patients (9).

Aims

These guidelines aim to provide updated recommendations for vaccination of egg-allergic individuals, consistent with international and current Australian and New Zealand Guidelines (10-13). Since the MMR vaccine contains no egg protein and egg-containing vaccines such as the Yellow Fever and Q Fever are specialist vaccines, this document will concentrate on the safety of administering the influenza vaccine in the egg allergic individual. The information applies to the vaccines currently available in Australia and New Zealand.

Since vaccine manufacture is subject to change and may vary between countries, this information may or may not be applicable to vaccines available in other countries. In drafting these guidelines, the authors have examined recent published studies of vaccination safety (Table 1) published international recommendations, product information sheets and relevant Australian and New Zealand vaccination guidelines. As with any form of medical intervention, the benefits of vaccination (protection against infection) need to be balanced against the very low risk of adverse reactions.

Minor short-lived side effects from vaccination are common

Administration site reactions (local pain, redness and swelling for intramuscular vaccines; nasal congestion, rhinorrhoea and sore throat for intranasal vaccines), fever, muscle aches, irritability or worsened eczema a day or so later are common vaccine side effects and do not represent vaccine allergy but rather the initiation of an immune response. Measles mumps and rubella (MMR) and Varicella vaccines can occasionally be followed by delayed rash (usually 4-12 days after vaccination) but this is also *not* due to a vaccine allergy. An exception to these may be a delayed (type 3) response characterised by urticaria and/or joint pain observed several hours to several days after vaccination. This type of reaction does not increase the risk of anaphylaxis and does not contraindicate subsequent vaccination.

Severe allergic reactions to vaccination are very rare

The risk of a severe allergic reaction to vaccination (anaphylaxis), to inactivated influenza vaccine is very low, estimated at 1.35 per million doses (11, 14). Immunisation guidelines recommend an observation period of 15 minutes (Australia) or 20 minutes (New Zealand) post-vaccination. The first symptoms of anaphylaxis may occur 20 minutes or longer after vaccination. However, because anaphylaxis is a very rare event it is not considered routine practice for vaccinated individuals to be observed beyond these specified times.

Vaccination providers should be able to recognise and manage anaphylaxis. Individuals suspected of having had anaphylaxis to vaccination should be referred for allergy specialist assessment before that specific vaccine is given again.

Most vaccines do not contain food allergens

There are no traces of dairy products, peanut, tree nuts, wheat, soy, seeds or seafood in vaccines. The following vaccines, as currently recommended on the Australian Childhood Vaccination Schedule (15) do not contain food-derived protein allergens and can be given to any patient with food allergy, even those with food-induced anaphylaxis:

- MMR (measles, mumps, rubella)*
- MMR-V (measles, mumps, rubella varicella)
- DPT (Diphtheria, Pertussis, Tetanus)
- IPV (Inactivated polio vaccine)
- HiB (*Hemophilus influenzae* type B)
- Pneumococcus
- Meningococcal C
- Chickenpox (Varicella) vaccines (which contain neither egg nor chicken protein)
- Rotaviral vaccine

* **Note:** The MMR vaccine is cultured on chicken fibroblast cell cultures, contains no residual egg allergen and has been safely administered to large numbers of egg-allergic individuals. The rare allergic reactions to MMR vaccination that have occurred have been attributed to non-egg ingredients such as gelatin (16).

The following vaccines may contain residual egg protein

- Seasonal inactivated and live attenuated influenza vaccines
- Pandemic inactivated influenza vaccine(s) (e.g. H1N1, bird or swine flu vaccines).
- Yellow Fever vaccine (important for travellers and those living in an endemic area)
- Q fever vaccine (important in occupational setting)

Egg allergy is common, with up to 8.9% of Australian infants having challenge proven allergy to raw egg and likely a smaller proportion reacting to well cooked egg (5). Most outgrow their allergy by primary school, although with occasional persistence or development of new egg allergy during adult life. Some vaccines are grown in eggs (see below).

Older vaccination guidelines recommended avoidance of influenza vaccination in egg-allergic individuals based on case reports of anaphylaxis 30 years ago when the amounts of egg protein were much higher than currently (6, 7).

To enhance safety, attempts have been made in recent years to limit the amount of egg ovalbumin in the pandemic, seasonal inactivated and live attenuated influenza vaccines to less than 1 microg of egg protein per vaccine dose.

The amount of residual egg protein in Yellow fever and Q fever vaccines is generally higher than in seasonal influenza and H1N1 vaccines, although administration of these vaccines to egg allergic individuals has been described (17).

Egg allergic patients in whom Yellow fever or Q fever vaccines are indicated should be referred to an allergy/immunology specialist for assessment. The remaining discussion pertains to egg-allergic individual in whom seasonal/pandemic influenza vaccination is indicated.

What is known about the safety of egg-containing influenza vaccines?

In the only case report of a death following influenza vaccine in 1969 (18), few details have been published, and the causative relationship with egg allergy is unclear.

More useful information has been obtained from recent prospective studies using H1N1 or seasonal influenza vaccines where the amount of residual egg ovalbumin has been limited to 1ug/dose or less.

A recent review of 28 studies comprising 4,315 subjects with egg allergy (including 656 subjects with a history of egg anaphylaxis) showed no severe reactions after influenza vaccination (8) as summarized in Table 1a (smaller older studies where the amount of ovalbumin is unknown are omitted).

Mild and occasional side-effects such as local itching, mild hives, throat irritation, wheezing or abdominal pain have been observed but not anaphylaxis. Comparison of graded administration of vaccination ("split dosing") has shown no differences in the rate of adverse reactions (Table 1a). Results of allergy testing with the vaccine prior to administration have shown no correlation with outcomes in terms of adverse reactions (19).

Similarly, collectively 955 children with egg allergy (including 412 with a history of egg-induced anaphylaxis) were shown to be safely vaccinated with LAIV without developing any immediate systemic reactions as summarised in Table 1b.

It is important to note that **the intranasal live attenuated influenza vaccine is contraindicated in people who are moderately and severely immunocompromised** (as for all live attenuated vaccines). People who are immunocompromised should consult with their treating specialist regarding suitable vaccine options.

Table 1a: Major studies of influenza vaccination of egg-allergic individuals: parenteral (injected) vaccines

Study	Ref	Location	No. participants	Outcome	Split/full dose	Design
Shimizu et al, Aregugi 2016	20	Japan	17	No systemic reactions	Split	Prospective
Gagnon et al 2010	21	Canada	830	No anaphylaxis; 17 had mild hives, wheeze or abdominal pain	Both	Prospective: 2+ years
Siret-Alatrissa et al 2011	22	France	178 (72 had +ve IgE)	No systemic reactions	Full	Prospective: 2-50 years
Des Roches et al, JACI 2012	23	Canada	367	No anaphylaxis; 1 case each of hives, vomiting or aggravated eczema	Both	Prospective;
Greenhawt et al, Annals Allergy 2012	24	USA	142	No systemic reactions	Both	Mixed randomized and retrospective observational
Webb et al, JACI 2011	25	USA	152	No systemic reactions	Both	Prospective; 1-30 years
Upton et al, Allergy Asthma Clin Immunol 2012	26	Canada	77	No systemic reactions	Split	Prospective: 0+ years
Forsdahl BMJ Open 2012	27	Norway	80	No systemic reactions	Both	Retrospective: 1- 17 years
Schuler et al. Can J Public Health 2011	28	Canada	62	No systemic reactions	Both	Prospective
James et al, J Ped 1998	29	USA	83	No systemic reactions	Both	Prospective
Fung I, Spergel JM. J Allergy Clin Immunol. 2012	30	USA	119 (all had egg-induced anaphylaxis)	No systemic reactions	Both	Retrospective:2007-2009
Tozandehjan et al, 2019	31	Iran	635 with egg allergy (cases) and 241 without egg allergy (controls)	No systemic reactions	Full	Case control

Table 1b: Studies of influenza vaccination of egg-allergic individuals: live attenuated intranasal vaccines (LAIV)

Study	Ref	Location	No. participants	Outcome	Split/full dose	Design
Turner et al, BMJ 2015	32	UK	779	No systemic reactions	Full; <i>intranasal vaccine</i>	Prospective: age 2-18 years
Turner et al, JACI 2015	33	UK	433	No systemic reactions	Full; <i>intranasal vaccine</i>	Prospective: age 2-17 years
Des Roches et al, JACI-IP 2015	34	Canada	69	No systemic reactions	Full; <i>intranasal vaccine</i>	Prospective: age 2-16 years.

International guidelines for vaccinating the egg-allergic individual

International guidance, including UK, USA, European and Canadian Consensus Guidelines, has evolved from recommending special precautions in egg-allergic individuals to affirming that egg allergy is not a contraindication to seasonal and pandemic influenza vaccination with standard modern products containing no more than 1 microg/dose of egg ovalbumin as summarised in Table 2.

Table 2. International guidelines for vaccinating the egg-allergic individual

Country	Organisation	Web link
Canada	NACI	https://www.canada.ca/en/public-health/services/immunization/national-advisory-committee-on-immunization-naci/canadian-immunization-guide-chapter-on-influenza-statement-on-seasonal-influenza-vaccine-2016-2017-addendum-laiv-use-egg-allergic-individuals.html
UK	BSACI	https://www.bsaci.org/bsaci-flu-vaccine-recommendations/
USA	CDC	https://www.cdc.gov/flu/prevent/egg-allergies.htm
Australia	Department of Health	https://www.health.gov.au/health-topics/immunisation/vaccines/influenza-flu-vaccine
New Zealand	Immunisation Advisory Centre	https://immune.org.nz/vaccine/influenza-vaccine
Other	World Allergy Organisation JTFPP Practice Parameter Update 2017	www.waojournal.biomedcentral.com/articles/10.1186/s40413-016-0120-5 https://iris.who.int/server/api/core/bitstreams/420b8e49-9f16-461d-a5d8-263069778432/content https://www.annallergy.org/article/S1081-1206(17)31187-0/fulltext

ASCIA Guidelines for vaccinating the egg-allergic individual

The Australasian Society of Clinical Immunology and Allergy (ASCIA) concurs with these views. Our recommendations specifically apply to vaccines containing no more than 1 microg ovalbumin per dose and are summarised below.

Recommendations

- Based on prospective and retrospective studies of influenza vaccination in those with and without egg allergy (including egg anaphylaxis), the presence of egg allergy does not increase the risk of allergic reactions to influenza vaccines. This includes both injectable inactivated, and intranasal live attenuated influenza vaccines.
- The entire vaccine can be administered in community vaccination clinics (which may or may not have direct medical practitioner supervision) as a single dose followed by the recommended 15 (Australia) or 20 (New Zealand) minute waiting period.
- In making this recommendation, we are aware that some guidelines (e.g. CDC 2017) recommend a longer waiting period of 30 minutes in those with past egg anaphylaxis and that occasionally allergic reactions to vaccination may commence later than 20-30 minutes after administration *The immediate availability of medical practitioner care is recommended and staff should be familiar with the recognition and treatment of anaphylaxis.*
- In individuals who have had suspected anaphylaxis following administration of the influenza vaccine itself, further vaccination should be avoided without specialist allergy assessment.
- If there is significant parental or health professional anxiety, the vaccine may be administered in primary care settings with a longer waiting period of 30 minutes.

Not recommended

- “Split dosing”
- Allergy testing with the vaccine or to egg prior to administration
- Ingestion of egg as a pre-condition to administering the vaccine (relevant in infants)
- Vaccination in specific hospital-based vaccination clinics
- Allergy specialist review before influenza vaccination unless anaphylaxis to the influenza vaccine itself has occurred previously

ASCIA Guidelines for vaccinating the egg-allergic individual and the Product Information (PI)

These guidelines are at variance with those contained in the Product Information (PI). In regard to egg allergy, ASCIA Guidelines and the Australian Immunisation Handbook should be followed and not the PI. This is only one example of a number of variations to product information that can be found in the Australian Immunisation Handbook.

If adverse reactions to vaccination occur

If a sudden collapse occurs after immunisation, a vaso-vagal event needs to be distinguished from anaphylaxis (see Australian Immunisation Handbook; <https://immunisationhandbook.health.gov.au/resources/tables/table-clinical-features-that-may-help-differentiate-between-a-vasovagal-episode-and-anaphylaxis>)

If the diagnosis is in doubt the vaccine recipient should be managed for anaphylaxis which includes the administration of IM/IN (intramuscular or intranasal) adrenaline (epinephrine).

Anaphylaxis management guidelines are on the ASCIA website <https://www.allergy.org.au/hp/anaphylaxis#aq3> and the Australian and New Zealand Committee on Resuscitation (ANZCOR) website <https://www.anzcor.org/home/first-aid/guideline-9-2-7-first-aid-management-of-anaphylaxis>

After the event, adverse events following immunisation should be reported to the Therapeutic Goods Administration (TGA) (<https://www.tga.gov.au/safety/report-problem/report-adverse-event-or-safety-problem>) in Australia and Medsafe in New Zealand (www.medsafe.govt.nz/).

It is important to document the timing of onset, the nature and severity of symptoms experienced, the likelihood of whether the adverse event occurred as a direct result of vaccination and details regarding underlying health issues (including known atopic disease). Blood should be taken for tryptase measurements if possible, to assist in the confirmation of possible anaphylaxis.

Conclusions

Current evidence is that patients with egg allergy (including anaphylaxis) for whom influenza vaccine is indicated can be vaccinated safely as long as the amount of residual egg ovalbumin is limited to 1microg or less per dose. This requires checking the egg ovalbumin content for any planned vaccine prior to administration, although at the time of writing (2026) all vaccines licensed for use in Australia and New Zealand contain less than 1microg egg ovalbumin.

Split dosing and prior allergy testing with the vaccine is no longer recommended.

Egg allergy does not increase the risk of anaphylaxis to the influenza vaccine but anaphylaxis to other components may occur. Vaccines should always be administered in facilities with staff able to recognise and treat anaphylaxis.

These recommendations are based on current available evidence, and subject to change as additional evidence becomes available.

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